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Governance logics and foresight functions: European strategic foresight in contrastive perspective

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Abstract

Foresight methods are often treated as transferable tools whose selection depends mainly on the object of analysis, the time horizon or the available data. This article argues that such a view is incomplete because foresight methods are embedded in governance settings and perform different institutional functions across them. The article does not claim that the context-dependence of foresight methods is new per se. Rather, it contributes by specifying this context-dependence through the lens of governance logics and by showing how the same foresight methods perform different institutional functions across technocratic, market-managerial, networked and anticipatory governance settings.

The purpose of the article is to develop a governance-sensitive framework that links foresight methods to institutional functions in public decision-making. Empirically, the article uses a contrastive case design. The European Union is treated as the focal case of institutionalised strategic foresight in a multi-level governance system, while Japan and South Korea serve as secondary contrastive cases. The design does not seek a fully symmetrical country-by-country comparison. Rather, it uses the Japanese and South Korean cases to contrast the EU trajectory with alternative governance settings in which foresight remains more strongly linked to technocratic priority-setting, developmental planning or data-driven policy adjustment. The analysis combines an exploratory bibliometric mapping of Scopus publications, VOSviewer-based semantic analysis of foresight-method terms, manual validation of 68 publications and qualitative case interpretation.

The findings suggest that foresight methods are not institutionally neutral. In technocratic settings, they mainly support expert prioritisation and cognitive opening; in market-managerial settings, strategic positioning and performance alignment; in networked governance, stakeholder alignment and legitimacy building; and in data-driven or anticipatory governance, early warning, resilience monitoring and policy adaptation. The EU case shows the most developed combination of networked and anticipatory functions. Strategic Foresight Reports, ESPAS, the EU-wide Foresight Network, the Better Regulation Toolbox, the European Green Deal, Open Strategic Autonomy and resilience dashboards together indicate that EU strategic foresight should be understood as a networked and anticipatory governance infrastructure rather than as a set of isolated forecasting techniques.

The article contributes to European futures research by shifting the discussion of foresight-method selection from universal methodological classifications to the institutional design of future-oriented governance. It also introduces

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the notion of function-sensitive foresight design as a complement to existing method taxonomies in futures studies.

Keywords Strategic foresight, Anticipatory governance, European futures research, Governance logics, Foresight methods, EU strategic foresight

Introduction

Foresight has become a key institutional practice through which public organisations deal with long-term uncertainty, technological change, social risks and alternative futures. Its role has expanded as governments and supranational institutions face challenges that cannot be addressed through short-term planning alone: climate transition, digitalisation, demographic change, technological sovereignty, resilient supply chains and the governance of research and innovation [29, 39, 64].

The European context makes these issues particularly salient. The European Union is a multi-level governance system in which long-term policy is shaped through interactions among EU institutions, member states, scientific communities, businesses, civil society and expert networks. Foresight in Europe therefore performs not only a forecasting function, but also coordinating, legitimising and anticipatory functions. It helps construct shared understandings of possible futures, align policy priorities, identify risks, support resilience and strengthen strategic autonomy [19, 28, 71].

Yet the selection of foresight methods in public policy is often described as a primarily technical task. Scenarios, Delphi, trend analysis, horizon scanning, stakeholder analysis, risk assessment, workshops and backcasting are frequently presented as tools whose suitability depends mainly on the object of analysis, the time horizon or data availability. This view is insufficient because foresight methods are not used in an institutional vacuum [44, 63].

This article builds on a line of research that has already questioned linear and method-centred understandings of foresight, technology assessment and innovation policy. Godin [30] showed the historical construction and limits of the linear model of innovation. Martin and Johnston [44] interpreted technology foresight as a way of wiring up national innovation systems, that is, as a mechanism that connects actors, priorities and institutions rather than an autonomous technique. Miles and Keenan [46] and Popper [60] demonstrated that foresight-method selection is multi-factorial and depends on objectives, process phases, forms of participation, scale, resource constraints and method combinations. Technology assessment (TA) literature develops an analogous argument for socio-technical assessment more broadly: methods are selected and combined in relation to problem framing, institutional mandate, stakeholder configuration, normative orientation and expected policy use [9, 10, 31].

Constructive technology assessment (CTA) is therefore used in this article not as the only relevant TA tradition, but as one strand within a broader TA and anticipatory-governance literature. CTA remains important because it emphasises reflexive interaction with ongoing innovation processes and the shaping role of expectations, scenarios and socio-technical imaginaries [61, 62, 66]. At the same time, future-oriented technology assessment connects foresight, anticipation and policy design more directly and shows why foresight should be analysed as part of assessment and governance rather than as an external methodological add-on [65]. Anticipatory governance extends this argument by linking foresight with engagement and integration as capacities for governing emerging technologies while governance choices can still shape their trajectories [4, 32].

Against this background, the article asks: how do different public-governance logics shape the institutional functions of foresight methods in strategic decision-making? The article does not claim that the context-dependence of foresight methods is new per se. Its contribution lies in specifying this dependence for public governance by showing how the same methods perform different institutional functions in technocratic, market-managerial, networked and anticipatory governance settings.

This article contributes to European futures research by shifting the discussion of foresight-method selection from universal methodological classifications to the institutional design of future-oriented governance. Rather than asking only which method should be selected for a particular foresight exercise, the article asks what institutional function a method performs in a specific governance setting. In doing so, it clarifies how strategic foresight becomes part of European anticipatory governance rather than a set of analytical techniques for forecasting or priority-setting.

The article develops a conceptual framework called governance logics–foresight functions. It distinguishes four governance logics: technocratic or hierarchical governance, market-managerial governance, networked or participatory governance and data-driven or anticipatory governance. Each logic creates a different demand for foresight: expert prioritisation, strategic positioning, stakeholder alignment, legitimacy building, early warning or adaptive monitoring [15, 36, 57, 59, 67].

Empirically, the European Union is treated as the focal case, while Japan and South Korea serve as contrastive cases. This formulation is deliberate. The article does not present a fully symmetrical comparatistic study of three foresight systems. The EU case is analysed in greater institutional depth because the article's main object is European strategic foresight in a multi-level governance system. Japan and South Korea are used more selectively to sharpen the analytical contrast: Japan illustrates a stronger technocratic and vision-oriented trajectory, while South Korea illustrates a developmental and increasingly data-driven trajectory [11, 12, 34, 52]. This contrastive design makes it possible to clarify what is specific about the EU case without overstating the empirical equivalence of the three cases.

Foresight methods and governance logics: theoretical background

Foresight as an institutional practice of future-oriented governance

Foresight is understood here not as a collection of forecasting techniques, but as an institutional practice for organising work with possible, plausible and desirable futures. Such a practice includes identifying trends and weak signals, constructing alternative scenarios, assessing technological priorities, engaging stakeholders, evaluating risks and building strategic orientations [29, 39, 64].

Historically, the evolution of foresight has involved both the expansion of methodological repertoires and the broadening of the actors involved in future-oriented work. Earlier approaches relied heavily on trend analysis, extrapolation, expert judgement and technology forecasting. Later generations incorporated Delphi, scenario planning, expert panels, interviews, SWOT analysis, horizon scanning, benchmarking, cost–benefit analysis, patent analysis, stakeholder analysis, workshops, risk management and backcasting [1, 13, 29, 33, 64].

For the present article, however, the key issue is not only the proliferation of methods but also the change in their institutional function. In contrast to studies that classify methods according to knowledge sources, phases of the foresight process or methodological combinations [46, 60], the unit of analysis here is the relation between method, function and governance logic. Foresight may support technological priority-setting, strategic positioning, stakeholder engagement, legitimacy building, shared visions, weak-signal monitoring or adaptive policy correction [8, 44, 63]. This functional perspective links foresight studies to critiques of linear innovation models and to broader technology assessment literature, including constructive technology assessment, future-oriented technology assessment and anticipatory governance [30–32, 61, 65, 66].

The broader TA perspective is useful because it has long addressed the context-dependence of method selection. TA does not treat indicators, scenarios, participatory formats or assessment tools as context-neutral instruments. It asks how analytical choices are shaped by the assessment problem, the institutional location of expertise, the degree of uncertainty, the actor constellation and the intended contribution to public decision-making. Böschen and Dewald [9] describe TA as contextualisation expertise, while Böschen et al. [10] show that even indicator work oscillates between context-neutralising and context-open strategies. This article transfers that insight to strategic foresight in public governance by specifying how governance logics condition the institutional functions of foresight methods.

From public administration models to governance logics

Public administration literature commonly distinguishes several broad models: classical bureaucracy, New Public Management (NPM), New Public Governance (NPG) or Good Governance, and more recent approaches based on data, indicators, risk assessment and anticipatory governance [36, 38, 57, 59, 67, 72].

A linear sequence of administrative models is not sufficient for futures research. First, governance models rarely replace one another completely. Bureaucratic, managerial, networked and data-driven elements often coexist within the same institutional system. Second, indicator-based or data-driven governance is better understood as an instrumental logic of data, indicators and monitoring than as an administrative paradigm equivalent to NPM or NPG [3, 15, 47].

The article therefore uses the concept of governance logics. This allows public governance to be analysed as a set of institutional logics that shape the role of foresight in different ways: technocratic or hierarchical, market-managerial, networked or participatory, and data-driven or anticipatory governance. This understanding is consistent with the view that foresight is part of an innovation and governance architecture. If Martin and Johnston [44] emphasised foresight as a mechanism for strengthening links within national innovation systems, the present article examines how governance logics change the function of methods: the same scenario, expert or participatory tool may support prioritisation, coordination, legitimacy or early warning depending on the setting in which it is embedded.

Institutional functions of foresight

The framework is based on six institutional functions of foresight. The first is cognitive opening: expanding actors' understandings of possible futures, challenging linear thinking and making uncertainty visible as a governance issue [41, 49].

The second is expert prioritisation: selecting scientific, technological and policy priorities through expert judgement, trend analysis, Delphi, expert panels and roadmaps

[29, 52, 69]. The third is strategic positioning: using foresight to identify competitive niches, assess market potential, define strategic advantages and align long-term objectives with performance criteria [81, 83].

The fourth is stakeholder alignment: involving actors, aligning expectations, identifying conflicts and distributing roles in long-term policy implementation [68, 82]. The fifth is legitimacy building: strengthening the political and social legitimacy of decisions through participation, public discussion of alternative futures and inclusion of different interests [21, 55]. The sixth is anticipatory adaptation: early warning, weak-signal monitoring, risk analysis and adaptive policy correction based on new evidence [19, 73, 80].

Conceptual framework: governance logics–foresight functions

The proposed framework links four public-governance logics to typical foresight functions and methods. It does not assign each method to one governance model. On the contrary, it assumes that the same method may perform different functions under different institutional conditions.

Table 1 summarises the proposed conceptual framework by linking governance logics to the future-oriented problems, institutional functions and foresight methods associated with them.

> P1. In technocratic or hierarchical governance, foresight mainly supports expert prioritisation and cognitive opening of planning.
> P2. In market-managerial governance, foresight methods support strategic positioning, competitiveness assessment and the alignment of long-term goals with performance criteria.
> P3. In networked or participatory governance, foresight shifts towards stakeholder engagement, interest alignment and the legitimacy of long-term decisions.
> P4. In data-driven or anticipatory governance, foresight becomes part of an infrastructure for early warning, weak-signal monitoring and adaptive policy correction.

Data and methods

Research design

The study uses a mixed-method design that combines quantitative mapping of literature with qualitative interpretation of institutional cases. The aim is not to test a statistical relationship between administrative models and foresight methods. Rather, the study identifies research associations and explains how foresight methods perform institutional functions in different governance settings. The case logic is contrastive rather than fully comparatistic: the EU is the focal case, while Japan and South Korea are used to highlight alternative ways in which foresight can be embedded in technocratic, developmental and data-driven governance trajectories.

The first stage was a literature review covering foresight studies, strategic foresight, technology foresight, future-oriented technology assessment, constructive technology assessment, anticipatory governance and public governance. The second stage was an exploratory bibliometric mapping of Scopus publications. This is an exploratory bibliometric mapping, not a systematic review. Each governance-related corpus combined a general foresight query with terms referring to bureaucracy, New Public Management, New Public Governance and indicator-based approaches. The third stage was a contrastive case analysis of the European Union, Japan and South Korea. The full search strings and validation decisions are provided in Additional file 1.

Bibliometric mapping

The bibliometric mapping used Scopus as the source database. The general foresight query included terms such as foresight, technology forecasting, future-oriented technology analysis, long-range planning,

Table 1 Governance logics and foresight functions

Governance logic	Key future-oriented governance problem	Main foresight function	Relevant methods
Technocratic/hierarchical governance	Rigid planning; dependence on expert bureaucracy; limited engagement with alternative futures	Cognitive opening; expert prioritisation	Delphi; expert panels; scenarios; trend analysis; interviews
Market-managerial governance	Efficiency; competitiveness; priority selection; performance assessment	Strategic positioning; performance alignment	SWOT; benchmarking; trend analysis; cost–benefit analysis; scenarios
Networked/participatory governance	Coordination of multiple actors; social risks; legitimacy of decisions	Stakeholder alignment; legitimacy building	Stakeholder analysis; workshops; participatory scenarios; backcasting; risk analysis
Data-driven/anticipatory governance	High uncertainty; weak signals; need for adaptive monitoring	Early warning; anticipatory adaptation	Horizon scanning; weak-signal analysis; indicators; dashboards; big data analytics; risk management

Source: Authors' elaboration based on Georghiou [29], Martin and Johnston [44], Saritas [64], Hood [36], Peters [59], Osborne [57], Dizdaroglu [15], Barben et al. [4], Guston [32], Böschen and Dewald [9], Böschen et al. [10], Grunwald [31], Scapolo et al. [65] and European Commission [19]

Table 2 Publication corpora used in the bibliometric analysis

Original corpus in the search query	Interpretation in the article	Number of Scopus records
Bureaucracy/Weberian model	Technocratic/hierarchical governance	96
New Public Management	Market-managerial governance	10
New Public Governance/Good Governance/collaborative governance	Networked/participatory governance	66
Indicator-based approach/indicator-based planning	Data-driven/anticipatory governance	18

Source: Authors' calculations based on Scopus. Search strings and access details are provided in Additional file 1

Table 3 Absolute frequency of foresight-method terms across publication corpora

Method term	Hierarchical/technocratic	Market-managerial	Networked/participatory	Data-driven/anticipatory
Scenario	22	2	37	3
Competitive intelligence	1	0	0	0
Horizon scanning	0	1	0	0
Workshop	2	0	4	0
SWOT analysis	3	4	0	0
Backcasting	0	0	2	0
Patent analysis	1	0	0	0
Benchmarking	1	0	1	1
Cost–benefit analysis	1	0	0	0
Trend analysis	8	7	5	5
Interview	13	1	5	0
Risk management	7	1	14	6
Stakeholder	13	1	20	5

Source: Authors' calculations based on Scopus corpora

Table 4 Relative frequency of foresight-method terms across publication corpora

Method term	Hierarchical/technocratic	Market-managerial	Networked/participatory	Data-driven/anticipatory
Scenario	23%	20%	56%	17%
Competitive intelligence	1%	0%	0%	0%
Horizon scanning	0%	10%	0%	0%
Workshop	2%	0%	6%	0%
SWOT analysis	3%	40%	0%	0%
Backcasting	0%	0%	3%	0%
Patent analysis	1%	0%	0%	0%
Benchmarking	1%	0%	2%	6%
Cost–benefit analysis	1%	0%	0%	0%
Trend analysis	8%	70%	8%	28%
Interview	14%	10%	8%	0%
Risk management	7%	10%	21%	33%
Stakeholder	14%	10%	30%	28%

Source: Authors' calculations. Percentages are calculated as the number of occurrences of a method term divided by the number of publications in the corresponding corpus

forward-looking analysis, strategic forecasting and visioning. This query was then crossed with four governance-related queries. The resulting publication corpora are presented in Table 2. Full search strings are provided in the supplementary materials.

The resulting corpora were analysed with VOSviewer. From the keywords and terms appearing in the publications, 13 foresight-method terms were extracted: scenario, competitive intelligence, horizon scanning, workshop, SWOT analysis, backcasting, patent analysis, benchmarking, cost–benefit analysis, trend analysis, interview, risk management and stakeholder analysis. Bibliometric mapping identifies research associations; it does not by itself prove that a method was actually used in a foresight exercise. For this reason, the interpretation is qualitative and theory-guided, and the frequency results are complemented by manual validation.

Tables 3 and 4 report the frequency results used for the exploratory bibliometric mapping: Table 3 presents absolute frequencies, while Table 4 provides the corresponding relative frequencies.

Table 4 reports the relative frequency of each foresight-method term within the corresponding publication corpus.

Manual validation of the publication corpus

Manual validation was conducted to reduce the risk of mechanically interpreting term frequencies. The unit of analysis was a single Scopus publication from the corresponding governance corpus. For corpora containing at least 20 publications, the 20 most cited records in the corresponding Scopus corpus were selected for manual validation. Records were sorted by the citation field in descending order; when citation counts were equal, the original export order was retained. For smaller corpora, all available records were validated. Validation used titles, abstracts, author keywords, index keywords, DOI and Scopus URL.

Each publication was coded according to four criteria: whether the identified term actually denoted a foresight method or future-oriented practice; which governance logic the publication addressed; what institutional function the method performed; and whether the publication should be included in the interpretation, included with a caveat or excluded. In total, 68 publications were validated: 20 in the technocratic or hierarchical corpus, 10 in the market-managerial corpus, 20 in the networked or participatory corpus and 18 in the data-driven or anticipatory corpus.

Limitations of the bibliometric stage

The bibliometric stage is exploratory. Term frequency does not demonstrate actual use of a method in a specific governance model; it indicates that a foresight-related term is associated with a particular governance vocabulary within the selected corpus. The corpus sizes also differ substantially. Results for New Public Management and indicator-based approaches require particular caution because these corpora are smaller than the bureaucracy and New Public Governance corpora. Quantitative results are therefore used to guide interpretation, not as stand-alone evidence. Substantive interpretation relies on manual validation and the comparative cases.

Bibliometric mapping of foresight methods

Scenario methods

Scenario methods appear in all four corpora and are especially prominent in the networked or participatory corpus. This suggests that scenarios should not be treated as a method with a fixed function. In technocratic governance, scenarios may help overcome the linearity of administrative planning and open a broader range of possible futures. In market-managerial settings, they support strategic positioning and environmental assessment. In networked governance, they become a medium for joint discussion, stakeholder alignment and shared visions. In data-driven or anticipatory governance, they may be integrated into monitoring, early warning and adaptive policy correction [41, 49, 70].

Trend analysis

Trend analysis is most frequent in the market-managerial corpus. This is consistent with the logic of New Public Management, where competitiveness, efficiency, strategic positioning and external assessment are central. In technocratic governance, trend analysis supports expert prioritisation. In networked governance, it provides a common language for coordination among public authorities, businesses, scientific organisations and society. In data-driven or anticipatory governance, trend analysis becomes part of indicator systems, dashboards and weak-signal monitoring [35, 82, 83].

Stakeholder analysis

Stakeholder analysis is particularly important for networked or participatory governance. Its frequency in this corpus indicates that participation is not an additional feature but a core element of networked foresight. In hierarchical settings, stakeholder analysis may identify resistance to change and clarify roles. In networked settings, it supports legitimacy and the alignment of expectations. In anticipatory governance, stakeholder inputs can be translated into indicators and monitoring arrangements [6, 68, 76].

Risk management

Risk management links foresight to uncertainty governance. In technocratic governance, it can help bureaucratic structures respond to change. In market-managerial governance, risk analysis is incorporated into performance management and assessment. In networked governance, risk analysis broadens to include public deliberation and long-term consequences. In anticipatory governance, it becomes part of early-warning infrastructure [5, 50, 55, 80].

SWOT, benchmarking and cost–benefit analysis

SWOT analysis, benchmarking and cost–benefit analysis are more strongly associated with market-managerial governance. Their function is to support strategic positioning, compare alternatives, assess competitiveness and justify decisions through performance criteria. Their limitations become visible when they are used without scenario or systems analysis: they are useful for assessing current positions and options, but weaker for deep uncertainty, social conflict and long-term transformation [27, 81, 83].

Horizon scanning and data-driven methods

Horizon scanning appears only weakly in the mapped corpus, which may reflect the search-query design and terminological diversity. Substantively, however, horizon scanning is central to data-driven and anticipatory governance. Its function is not to produce a single forecast, but to support continuous observation of the changing environment, identify weak signals and enable timely policy adjustment [8, 19].

Manual validation of the publication corpus

Manual validation was conducted to check the substantive meaning of the bibliometric results. In total, 68 publications were assessed: 20 in the technocratic or hierarchical corpus, 10 in the market-managerial corpus, 20 in the networked or participatory corpus and 18 in the data-driven or anticipatory corpus. Each record was coded for whether the identified term denoted a foresight method or practice, which governance logic it addressed and which institutional function the method performed.

The validation shows that automated term detection provides useful research associations, but not every occurrence of scenarios, stakeholders, risk or long-range planning refers to foresight as a method. Interpretation therefore relies on the validated institutional function of the method, not on frequency alone.

Table 5 summarises the manual validation results and distinguishes between fully relevant, partially relevant and non-relevant records in each corpus.

The technocratic or hierarchical corpus contained the highest share of ambiguous cases. Only four of 20 publications directly confirmed the use of foresight methods, ten were partially relevant and six were excluded. This suggests that long-range planning or modelling in bureaucratic contexts is often broader than foresight. Nevertheless, validated publications support the function of cognitive opening: Neumann and Overland [49] use perspectivist scenarios to move policy planning beyond a linear view of the future; Burrows and Gnad [8] describe strategic foresight as a way to regain political initiative,and Constantinides [14] shows the importance of institutional receptivity to weak signals in crisis management.

In the market-managerial corpus, five of ten publications were fully relevant and four partially relevant. This corpus confirms links between foresight, strategic positioning, innovation governance and performance alignment. Zechlin [81] connects strategic planning with organisational priorities; Zielinski et al. [83] use technological trends as a basis for innovation governance to 2030; and El-Ghalayini [16] shows how digital transformation may become a tool of anticipatory governance.

The strongest connection was found between foresight and networked or participatory governance. Nine of 20 publications were directly relevant and eight partially relevant. Here foresight methods are used for stakeholder alignment, shared visions and legitimacy building. This is shown by backcasting in the bioeconomy [68], scenario-based strategic planning [70], public deliberation on risks [55] and EU research and innovation foresight linked to grand societal challenges [7].

In the data-driven or anticipatory corpus, eight of 18 publications were fully relevant and five partially relevant. The corpus confirms a shift towards indicators, monitoring, risk assessment and early warning. Wilson et al. [73] discuss national indicators for monitoring biological invasions; Asbridge et al. [2] link coastal flood risk assessment to vulnerability indicators; Ben-Daoud et al. [6] translate stakeholder actions into water-management indicators; and Wilts and Britz [74] connect scenarios, indicators and the long-term assessment of sustainability pathways.

Manual validation therefore clarifies the limits of bibliometric interpretation. Frequency analysis is useful for exploratory mapping, but a meaningful conclusion requires checking whether the term denotes a foresight method and what function it performs. This is especially important for the EU case, where the same methods simultaneously support coordination, legitimacy building, resilience monitoring and anticipatory adaptation.

European strategic foresight in contrastive perspective

European Union: networked and anticipatory foresight

The European Union is the focal case because strategic foresight has been institutionalised most explicitly there as a mechanism of multi-level, networked and anticipatory governance. Unlike national systems in which foresight is often embedded in vertical technological priority-setting, EU foresight operates under distributed competences among EU institutions, member states, expert networks, research organisations, businesses and civil society. Its function therefore extends beyond technology forecasting. It supports policy coordination, long-term orientation, risk analysis, resilience and Open Strategic Autonomy [19, 20, 28].

Table 5 Summary of manual validation results

Original corpus	Governance logic	Validated	Yes	Partial	No	Exclude
Bureaucracy/hierarchical logic	Technocratic/hierarchical governance	20	4	10	6	8
Indicator-based/data-driven logic	Data-driven/anticipatory governance	18	8	5	5	6
New Public Management/managerial logic	Market-managerial governance	10	5	4	1	1
New Public Governance/networked logic	Networked/participatory governance	20	9	8	3	4

Source: Authors' manual validation of 68 Scopus publications. "Yes" indicates that the term denotes a foresight method or practice; "Partial" indicates a functionally related practice or indirect link; "No" indicates non-relevance

A key point of institutionalisation was the first annual Strategic Foresight Report in 2020. The Commission framed strategic foresight as part of EU policy-making and linked it to resilience after the COVID-19 crisis. The report analysed resilience across four interconnected dimensions—social and economic, geopolitical, green and digital—and proposed resilience dashboards as a tool for assessing vulnerabilities and capacities in the EU and its member states [19]. From that point, strategic foresight should be viewed not as an isolated expert exercise but as part of the policy cycle, connected to the preparation of initiatives, impact assessment and the adjustment of political priorities.

The annual Strategic Foresight Reports form a sequence of anticipatory governance. The 2021 report connected foresight with Open Strategic Autonomy and the EU's capacity and freedom to act under long-term climate, technological, geopolitical and demographic shifts [20]. The 2022 report addressed the twinning of the green and digital transitions in a new geopolitical context, showing that digital technologies may accelerate ecological transformation while also creating resource, energy and regulatory risks [22]. The 2023 report focused on sustainability and people's wellbeing as the core of Europe's Open Strategic Autonomy, connecting environmental, social and economic dimensions of transition [23]. The 2025 report developed the notion of Resilience 2.0 as a proactive and forward-looking approach extending to 2040 and beyond [25].

A second element of the European model is the European Strategy and Policy Analysis System (ESPAS). ESPAS is an interinstitutional cooperation mechanism involving the European Commission, the European Parliament, the Council of the EU, the European External Action Service and EU advisory bodies. It monitors global trends and provides strategic foresight for EU decision-makers. ESPAS produces Global Trends Reports, horizon-scanning reports and foresight papers, thereby supporting the circulation of strategic knowledge among EU institutions [17, 24]. In the proposed framework, ESPAS performs a function of coordination and anticipatory intelligence: it does not merely produce knowledge about futures but embeds it in interinstitutional interaction.

A third element is the EU-wide Foresight Network, including the Ministers for the Future and a network of senior officials from national administrations. This mechanism enables exchange between the foresight capacities of the Commission and those of the member states, providing a continuing forum for discussing Europe's future challenges [24]. Future-oriented work in the EU is therefore organised not only through analytical reports but also through inter-administrative coordination. This strengthens the networked character of European foresight by creating a shared strategic attention space across governance levels.

A fourth element is the integration of foresight into the Better Regulation framework. The Better Regulation Toolbox includes Tool #20, "Strategic foresight for impact assessments and evaluations," which means that foresight is used not only in long-term strategies but also in impact assessments, evaluations and the development of policy options [26]. This is important because scenarios, horizon scanning and weak-signal analysis are inserted into the procedural architecture of decision preparation. They help stress-test policy options and align regulation with future risks and opportunities.

The European Green Deal and Open Strategic Autonomy provide the substantive policy agenda within which strategic foresight acquires a transformational function. The Green Deal defines a long-term transition towards a resource-efficient and competitive economy, climate neutrality by 2050, and coordination across energy, industry, transport, agriculture, finance, labour markets and regional development [18]. Open Strategic Autonomy links the EU's capacity to act with climate, technological, geopolitical and social changes [20, 23]. In these policy frames, foresight is used not only to forecast an external environment but also to identify trade-offs among climate, digital, social and geopolitical objectives.

Resilience dashboards are the most concrete instrument of data-driven and anticipatory governance in the EU case. Proposed in the 2020 Strategic Foresight Report and developed by the Joint Research Centre, they function as a forward-looking monitoring tool for assessing vulnerabilities and capacities across social and economic, green, digital and geopolitical dimensions [19, 40]. The dashboards translate foresight from the format of a periodic report into a monitoring system that links qualitative foresight, indicators, vulnerability assessment and policy action.

EU strategic foresight should therefore be understood as a networked and anticipatory governance infrastructure rather than as a set of isolated forecasting techniques. Strategic Foresight Reports establish long-term policy frames; ESPAS and the EU-wide Foresight Network support interinstitutional and inter-administrative coordination; the Better Regulation Toolbox embeds foresight in impact assessment and evaluation; the European Green Deal and Open Strategic Autonomy connect futures work to transformative policy agendas; and resilience dashboards translate foresight into indicator-based monitoring. This institutional configuration distinguishes the EU from more technocratic models of Japanese foresight and from South Korea's more developmental and data-driven trajectory.

Table 6 Contrastive characteristics of institutional foresight functions

Parameter	European Union	Japan	South Korea
Status in the article	Focal case	Contrastive case	Contrastive case
Dominant governance logic	Networked/participatory—> anticipatory governance	Technocratic—> anticipatory governance	Developmental—> market-managerial—> data-driven governance
Main foresight function	Coordination, legitimacy, resilience, early warning	Expert prioritisation, long-term visioning	Technological positioning, roadmapping, adaptive correction
Main methods	Participatory scenarios, stakeholder analysis, horizon scanning, risk analysis	Delphi, expert panels, scenarios, trend analysis, patent analysis	Delphi, roadmapping, trend analysis, benchmarking, patent analysis, big data
Policy linkage	Strategic Foresight Reports, research and innovation policy, Green Deal, strategic autonomy agenda	Science and Technology Basic Plans, Society 5.0	Basic S&T Plans, digital government strategies, technology roadmaps
Specificity	Foresight as infrastructure of multi-level European governance	Foresight as a tool of technocratic visioning	Foresight as a tool of developmental and data-driven policy adjustment

Source: Authors' elaboration based on European Commission [19, 20], Gaiti and Georgakakis [11, 28, 52, 54], Choi and Choi [12], Heo and Seo [34], NIA [51] and WIPO [75]

Japan: technocratic foresight and long-term visioning

Japan provides a contrastive case. Its foresight system developed for a long period under strong state coordination, expert bureaucracy and technological priority-setting. After the Second World War, Japanese public governance focused on reconstruction, industrial development and technological sovereignty. In this configuration, foresight mainly performed the function of expert prioritisation [52, 77, 78].

NISTEP technology foresight exercises, based on expert surveys and Delphi methods, played a central role. Their purpose was to identify long-term science and technology directions capable of supporting industrial development and state objectives. In the proposed framework, this corresponds to technocratic or hierarchical governance: foresight is used for expert selection, long-term planning and cognitive opening of possible technological trajectories [41, 54].

Compared with the EU, where foresight has increasingly become a mechanism of multi-level coordination and participation, the Japanese trajectory remained more technocratic for longer. The recent Society 5.0 agenda has broadened the function of foresight. It is no longer only about technology forecasting but also about constructing an integrated vision of a future society in which digital technologies, artificial intelligence, robotics and data address social problems [11, 45, 53]. Japan shows that anticipatory governance can develop within a system with strong state coordination, although the participatory and legitimising functions are less pronounced than in the EU.

South Korea: developmental planning and data-driven foresight

South Korea illustrates a transition from developmental planning to data-driven strategic foresight. Early technology forecasting and foresight were embedded in state-led industrialisation, catching-up development and R&D priority-setting. The state played a central role in defining industrial directions, mobilising resources and coordinating major economic actors [12, 43, 48, 56].

Early foresight practices therefore supported expert and strategic prioritisation. During reforms associated with New Public Management, foresight increasingly performed the function of strategic positioning: technology forecasts, roadmapping, benchmarking, trend analysis and expert surveys were used to assess competitiveness, commercialise innovation, select R&D priorities and connect state programmes with performance [12, 58, 79].

In the contemporary period, digital government and AI-related instruments have reinforced the data-driven component. Big data analysis, patent analytics, digital platforms and dynamic adjustment of technology roadmaps make foresight part of adaptive governance. Its function shifts from periodic forecasting towards the continual correction of strategic priorities on the basis of new data [37, 42, 51, 75]. South Korea thus shows how foresight can be embedded in performance-oriented and data-driven governance, with a stronger emphasis on technological competitiveness and adaptive policy adjustment than on multi-level legitimacy building.

Contrastive analysis

Table 6 contrasts how foresight methods are embedded in different institutional functions across the focal EU case and the two contrastive cases.

The contrast indicates that differences between foresight systems are not determined only by the methods used. They also depend on the institutional functions those methods perform. Delphi, scenarios, trend analysis, stakeholder analysis and dashboards can be embedded in expert planning, strategic positioning, actor coordination, legitimacy building or adaptive monitoring.

The EU displays the most developed combination of networked and anticipatory functions: foresight supports multi-level coordination, public engagement, resilience analysis and Open Strategic Autonomy. Japan retains a more technocratic profile oriented towards expert science and technology priority-setting and long-term visioning. South Korea links foresight more closely to developmental planning, technology roadmapping and digital adaptation. These cases are not presented as empirically equivalent units of a full comparative design; they are used to make visible the institutional specificity of the European case.

Discussion

Foresight methods as institutionally embedded practices

The findings indicate that the institutional setting changes the meaning of foresight methods. Scenarios, stakeholder analysis, risk analysis and indicators do not have one fixed function. Depending on the governance logic in which they are embedded, they may support cognitive opening, strategic positioning, stakeholder alignment, legitimacy building or anticipatory adaptation. The analytical unit of this article is therefore not the method in isolation, but the relation between method, function and governance logic.

From method selection to function-sensitive foresight design

The proposed framework does not replace existing method taxonomies in futures studies or the broader methodological repertoire of technology assessment. Classifications such as the Foresight Diamond remain important for understanding the knowledge sources, methodological orientation and participatory intensity of foresight methods. TA literature adds a complementary insight: assessment methods are always embedded in institutional contexts, actor constellations and normative choices. The governance logics–foresight functions framework builds on this insight by adding a public-governance dimension. It explains why the same method may support expert prioritisation, strategic positioning, stakeholder alignment, legitimacy building or anticipatory adaptation depending on the governance logic in which it is embedded.

For futures studies and technology assessment, this implies a shift from method-based to function-sensitive foresight design. If Miles, Keenan and Popper explain method selection through process phase, method type, scale, participation and exercise context, the proposed framework adds the question of governance function: should the method support expert prioritisation, strategic positioning, stakeholder alignment, legitimacy, monitoring or anticipatory adaptation? This is close to TA debates on context-sensitive assessment, but it specifies the issue for strategic foresight in public governance. In practical terms, foresight design should start not only with the question of which tool to use, but also with the institutional task that foresight must perform in a given governance system [9, 10, 46, 60, 65].

The European trajectory of strategic foresight

The EU case reveals a particularly complex combination of functions. Strategic Foresight Reports provide a long-term policy frame; ESPAS and the EU-wide Foresight Network support interinstitutional coordination; the Better Regulation Toolbox embeds foresight in impact assessment and evaluation; the Green Deal and Open Strategic Autonomy set a transformational agenda; and resilience dashboards translate foresight into a system for monitoring vulnerabilities and capacities. The EU therefore differs from Japan and South Korea not only through stronger participation, but through a more developed combination of networked governance and anticipatory infrastructure [19, 20, 24, 40].

Practical implications for foresight process design

The framework can guide the design of public foresight and technology-assessment processes. In hierarchical systems, priority should be given to methods that expand the space of possible decisions. In managerial systems, tools for comparing alternatives and long-term risks are more relevant. In networked governance, participatory scenarios, stakeholder analysis and backcasting are central. In data-driven or anticipatory governance, horizon scanning, indicators, dashboards and risk monitoring should become elements of a continuous adaptation cycle. The broader TA literature helps avoid treating these tools as neutral techniques: their use should be justified by the governance problem, the actor constellation and the expected policy function [8, 31, 55, 80].

Limitations and future research

The study has several limitations. First, the bibliometric analysis is exploratory. Term frequencies do not prove actual method use; they indicate research associations between governance vocabularies and foresight-related terms.

Second, the corpus sizes differ. Results for market-managerial governance and data-driven or anticipatory governance should be interpreted cautiously because these corpora are smaller than the hierarchical and networked governance corpora.

Third, the cases of the EU, Japan and South Korea represent distinct institutional trajectories but do not exhaust the possible variety of foresight systems. Moreover, the study uses Japan and South Korea as contrastive cases, not as fully symmetrical empirical cases. Future research could extend the analysis to the United States, China, the United Kingdom, Nordic countries and individual EU member states, using a more balanced comparative design if the research objective is direct cross-country comparison.

Fourth, further studies could manually code larger corpora of government strategies, foresight reports and research-policy documents. This would allow researchers to move from bibliometric mapping of academic publications to a more direct analysis of how foresight methods are institutionalised in public governance.

Conclusion

The article has argued that foresight methods should be analysed as institutionally embedded instruments for working with futures. Their meaning is shaped not only by methodological procedure but also by the governance logic in which they are used: technocratic or hierarchical, market-managerial, networked or participatory, or data-driven and anticipatory.

The European Union illustrates the most developed shift towards networked and anticipatory foresight. In the EU case, foresight is institutionalised as an infrastructure for multi-level coordination, public participation, resilience monitoring and strategic orientation. In Japan, foresight remains more closely linked to technocratic science and technology priority-setting; in South Korea, it is more strongly connected to developmental planning, roadmapping and data-driven policy adjustment.

The article's contribution is not the claim that foresight methods depend on context; that point is already present in studies of method selection, innovation systems, technology assessment, constructive technology assessment, future-oriented technology assessment and anticipatory governance. Rather, the contribution lies in specifying this dependence through the governance logics–foresight functions framework. The framework explains why the same foresight method performs different tasks across institutional contexts, while manual validation of 68 publications shows that such differences cannot be inferred from term frequencies without checking the function that the method performs. This is particularly important for European strategic foresight, where future-oriented methods simultaneously support coordination, legitimacy, resilience and anticipatory adaptation.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s40309-026-00278-8.

Additional file 1: Supplementary bibliometric and validation data. Description: The file contains Scopus search strings, publication counts, absolute and relative frequency tables of foresight-method terms, VOSviewer terms, the validation coding scheme, the manual validation corpus and the validation summary

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Declarations

Competing interests
The authors declare that they have no competing interests.

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